

BRENDAN HARRIS

Postdoctoral Research Associate (2026–)

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RESEARCH THEMES

I study the principles of neural computation using statistical physics, dynamical systems, new analytical methods, and large-scale data:

- **Spatiotemporal dynamics:** how nested traveling waves coordinate hierarchical neural processing and guide flexible behavior.
- **Criticality:** noise-robust methods to track the distance to criticality in noisy, poorly sampled systems.
- **Stochastic processes:** fractional mean-field theory for neural circuits, fractional diffusion for decision-making and visual sampling.
- **Population geometry:** low-dimensional coding and communication subspaces, new methods for dimensionality reduction.

PUBLICATION RECORD

6 peer-reviewed publications, 56 citations, h-index 4. 2 first-author papers in Q1 journals (*Physical Review X*, top 3% General Physics and Astronomy, Scopus CiteScore; *Nature Communications*, top 3% Multidisciplinary). All first-author work accompanied by [open-source code](#).

EDUCATION

2022–2026	Physics PhD “Cross-scale dynamics in the working regime of the visual cortex” <i>Supervisor:</i> Prof. Pulin Gong. <i>Chapters:</i> • “Tracking the distance to criticality in systems with unknown noise”: Physical Review X (2024) • “Nested spatiotemporal theta–gamma waves organize hierarchical processing across the mouse visual cortex”: Nature Communications (2026) • “Anomalous dynamics in the working regime of the visual cortex”: In preparation (slides)	The University of Sydney
2021	Physics Honours, Class I and the University Medal “Inferring parametric variation across non-stationary time series” <i>Supervisor:</i> A/Prof. Ben Fulcher.	The University of Sydney
2018–2020	Bachelor of Science / Bachelor of Advanced Studies (Dalyell Scholar) Majored in Physics and Neuroscience.	The University of Sydney

LEADERSHIP, SERVICE AND COMMUNITY ENGAGEMENT

2024–	Co-organizer, Systems Neuroscience and Complexity seminar series Organize a cross-disciplinary seminar series across physics, medicine, and computer science	The University of Sydney
2018–	Open-source software developer and maintainer Develop and maintain Julia packages for time-series analysis and scientific research, enabling reproducible neural-data workflows; released repositories have attracted 80+ GitHub stars.	GitHub: brendanjohnharris
2022–	Open-science workshops Pre-COSYNE BrainHack (MILA, 2025); BrainHack Australia 2022; OpenScope NeuroDataReHack (Allen Institute, 2022).	Sydney; Montreal; Seattle
2018–	Public communication Authored a public-facing article in The Conversation with an accompanying video summary; ongoing science communication via X, Bluesky, GitHub, and personal website.	Sydney

INDUSTRY EXPERIENCE

2023	Neurofeedback software developer Developed software to interface with EEG headsets, infer brain-network activation, and implement neurofeedback for depression monitoring and treatment.	Resonait Medical Technologies
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SUPERVISION AND TEACHING

2025–	Secondary supervision of Physics Honours students • Jude Metcalf (2026) — <i>Adaptive fractional dynamics of visual sampling</i> • Aiden Sloots (2025) — <i>Bi-fractional diffusion for decision-making dynamics</i>	The University of Sydney
2022–2024	Lab tutor: interdisciplinary, computational, and statistical physics	The University of Sydney

CONFERENCE PRESENTATIONS

2025	OHBM 2025, Brisbane	Poster
2025	EPC/APCV 2025, UNSW	Talk
2025	COSYNE 2025, Montreal	Poster
2025	NeuroEng 2025, The University of Melbourne	Talk
2023	IBRO World Congress 2023, Granada	Poster
2022	31st Annual Computational Neuroscience Meeting (CNS*2022), Melbourne	Poster & workshop talk
2022	28th Annual Meeting of the Organization for Human Brain Mapping (OHBM)	Virtual poster

SELECTED TALKS

2026	Sydney – Tokyo: Neural and Nonlinear Dynamics Workshop	Talk
2026	Maths in the Brain (Melbourne)	Talk
2025	Center for Neuroscience Imaging Research (CNIR), IBS, South Korea	Invited talk
2024	Emerging Aspirations in Complex Systems, The University of Sydney	Workshop talk
2024	School of Physics HDR symposium, The University of Sydney	Workshop talk
2023	MIP:Lab, Geneva Biotech Campus (host: Dr. E. Amico)	Hosted visit
2023	Cognitive Neuroscience Hub, The University of Melbourne (host: Dr. J. Paul)	Hosted visit
2023	tLab, Monash University (host: Prof. N. Tsuchiya)	Hosted visit
2023	Complexity, Criticality & Computation Symposium (C3-2023), Heron Island	Workshop talk
2022	Yale Medical School (host: Dr. E. Lake)	Invited talk
2022	Emerging Aspirations in Complex Systems, The University of Sydney	Workshop talk

WORKSHOPS

2025	Pre-COSYNE BrainHack 2025, Montreal	Workshop
2024	Strong Compute Chess AI Hackathon, Sydney	Workshop
2022	BrainHack Australia 2022, The University of Sydney	Workshop
2022	OpenScope NeuroDataReHack, Allen Institute, Seattle	Workshop

SELECTED AWARDS

2026	Best ECR talk prize, Maths in the Brain 2026	Prize
2025	COSYNE Presenters Travel Grant	Travel
2023	Postgraduate Research Support Scheme	Travel
2023	Emerging Aspirations in Complex Systems Award	Travel
2022	Research Training Program (fee offset and stipend)	PhD scholarship
2022	Travel support for OpenScope NeuroDataReHack (Allen Institute)	Travel
2022	University Medal and Honours Class I; Dean's List of Excellence	Recognition
2021	Physics Foundation Scholarship No. III; Academic Merit Prize	Scholarships

REFEREES

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